Willem Donnelly

Data 4000-01 - Python Programming with AI

Professor Mowerman

Due: 11/11/26

Project Title: Fourth & Data: NFL Analytics for Fantasy and Betting

Description: Fourth & Data will be a Streamlit web app that collects NFL statistics in one place, including niche but highly predictive metrics that most fantasy and betting sites leave out or charge for, such as target share, air yards share, snap share, red zone opportunities, and expected points added (EPA). Users will be able to look up players and teams, compare players side by side, and see how a player's recent performance compares to a betting line or fantasy projection. The app will be built in Python using functions, loops, and object-oriented programming (classes such as Player and Team), with public NFL data from the nflverse project. It is an analytics and research tool and does not place bets.

Intended Users: Fantasy football players who want an edge in lineup, waiver, and trade decisions, and sports bettors (21+) who research player props and moneylines before making their own decisions.

Planned Features:

1. Player lookup with season and week-by-week stats, including usage metrics like target share, snap share, and air yards
2. Side-by-side player comparison for start/sit and trade decisions
3. Player prop analyzer that shows how often a player has gone over a given line this season and in recent games
4. Matchup view showing how each defense performs against each position
5. Team moneyline view comparing offensive and defensive efficiency